

Rohan Bajpai

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EDUCATION

University of Maryland

College Park, MD

Bachelor of Science in Computer Science

August 2021 – December 2024

- **GPA:** 3.7
- **Honors:** CS Departmental Honors, Presidential Scholarship Recipient, National Merit Finalist, Dean's List
- **Relevant Coursework:** Algorithms, Compilers, Applied Linear Algebra, Applied Probability & Statistics, Introduction to Parallel Computing, Introduction to Machine Learning
- **Clubs and Organizations:** Software Engineer at Hack4Impact, Webmaster at Phi Chi Theta

TECHNICAL SKILLS

Languages: Kotlin, Java, Python, C/C++, HTML, JavaScript, SASS, CSS

Technologies: Flask, Spring Boot, Node.js, Django, Elastic, Gym, PySpark, Pandas, MongoDB, React, Kafka

Tools: Git, Docker, GitLab, GitHub, AWS (Certified Developer)

EXPERIENCE

Software Development Engineer Intern

May 2023 – August 2023, May 2024 - Present

Amazon Web Services

Arlington, VA

- Framed a machine learning classification problem to improve data quality, significantly enhancing core infrastructure reliability through preemptive maintenance
- Automated MySQL database credential rotation via API, eliminating a **10+** day manual process, significantly improving security and team operational efficiency
- Implemented SSO authentication system, reducing beta testing onboarding time by **90%** for new customers
- Exceeded internship goals by completing primary project, stretch goal, an additional project, and machine learning training within a 12-week timeframe

Software Engineer Intern - Machine Learning

September 2023 – Present

Capital One

College Park, MD

- Deployed a distributed real-time model predicting account-customer mappings with sub-second latency and **99.99%** uptime
- Generated **2 million** synthetic data points using statistical sampling and reverse feature engineering for performance testing
- Optimized **DynamoDB** keys based on data access patterns, reducing system latency and enhancing overall performance

Software Engineering Co-op

June 2022 – May 2023

Leidos

Reston, VA

- Developed full-stack web app and API for predictive analytics visualization with **Spring Boot, PostgreSQL Thymeleaf, HTML, Bootstrap, and CSS** on a team of seven
- Presented new features, system designs, and updates to external customers in monthly progress meetings
- Proposed and engineered single-sign-on with secured access between application and **Elasticsearch/Kibana** to enhance user experience based on customer feedback
- Applied innovative internal AI/ML research to define requirements and design an RL-based automated firewall hardening product using **Python, Gym, and stable-baselines-3**
- Setup automated **CI/CD** pipeline within **GitLab** to lint then generate PDFs from markdown documentation

PROJECTS

Plato | Python, AWS, LangChain, ChromaDB

December 2023 – Present

- Launched a product enabling engineers to seamlessly search, understand, and integrate internal microservices
- Implemented novel **RAG** and re-ranking techniques, enhancing product performance and reducing costs

Melp | Python, Flask, React Native, Node.js, MongoDB

Spring 2022 – December 2022

- Developed a MERN stack mobile app for roommates, featuring bill-splitting, location-sharing, messaging, and chore tracking.
- Created a **Flask**-based microservice using **Tesseract** for OCR and **OpenCV** for image preprocessing, improving OCR results with a GPT-3 fine-tuned model for data normalization